Unit 2: Data Warehouse Logical Design

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Outline

- DBMS vs. Data warehouse
- Data marts
- Meta data
- Multidimensional data model
- Data cubes
- Schemas for Multidimensional model

DBMS vs. Data warehouse

Category	Data Warehouse	Database
Purpose	Analysis, Decision making	Day to day use
Support For	OLAP(on-line analytical processing)	OLTP(on-line transaction processing)
Data model	Multidimensional	Rational
Age of data	Current & time series	Current & real time
Data modification	Read/access only	Insert, update, delete
Type of data	Static	Dynamic
Amount of data per transaction	Larger	Smaller
Schema design	Renormalizations	Normalization

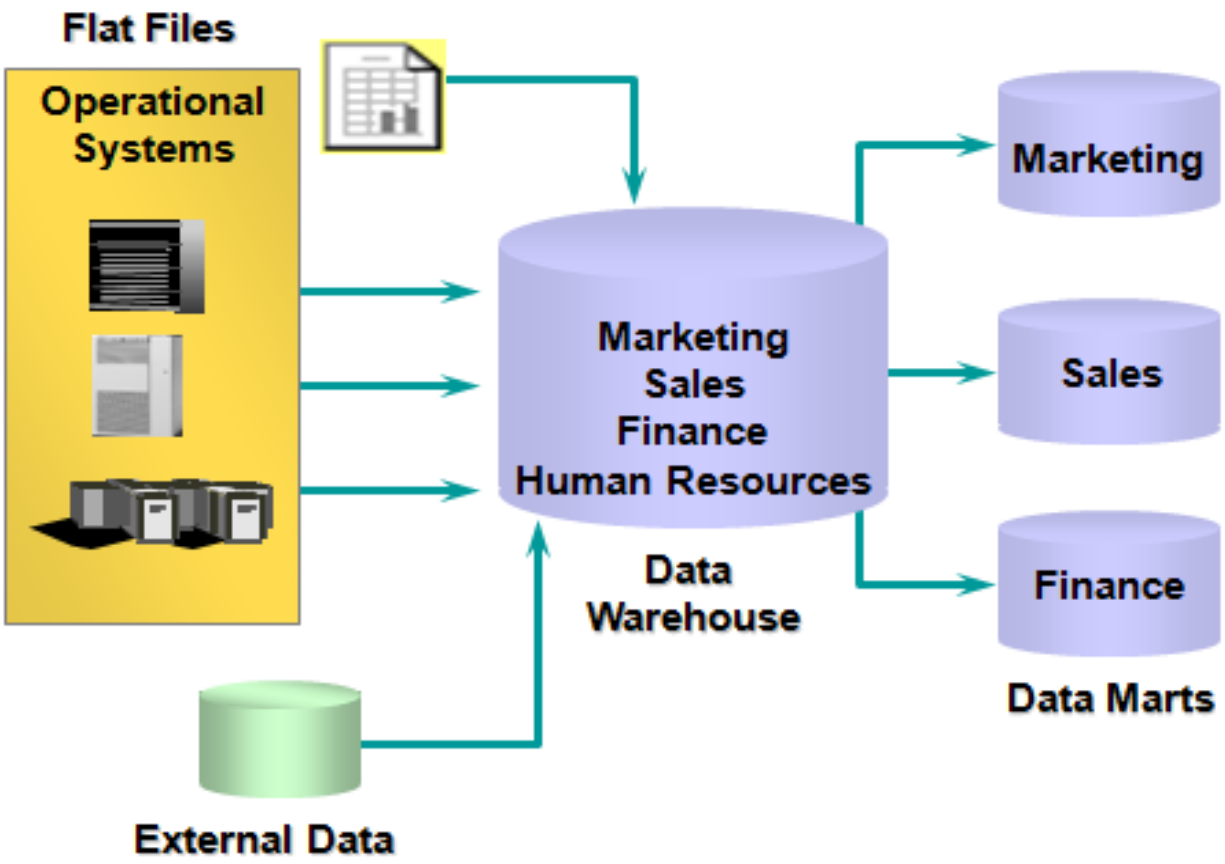
Data Marts

"A *data mart* is a subset or an aggregation of the data stored to a primary data warehouse. It includes a set of information pieces relevant to a specific business area, corporate department, or category of users".

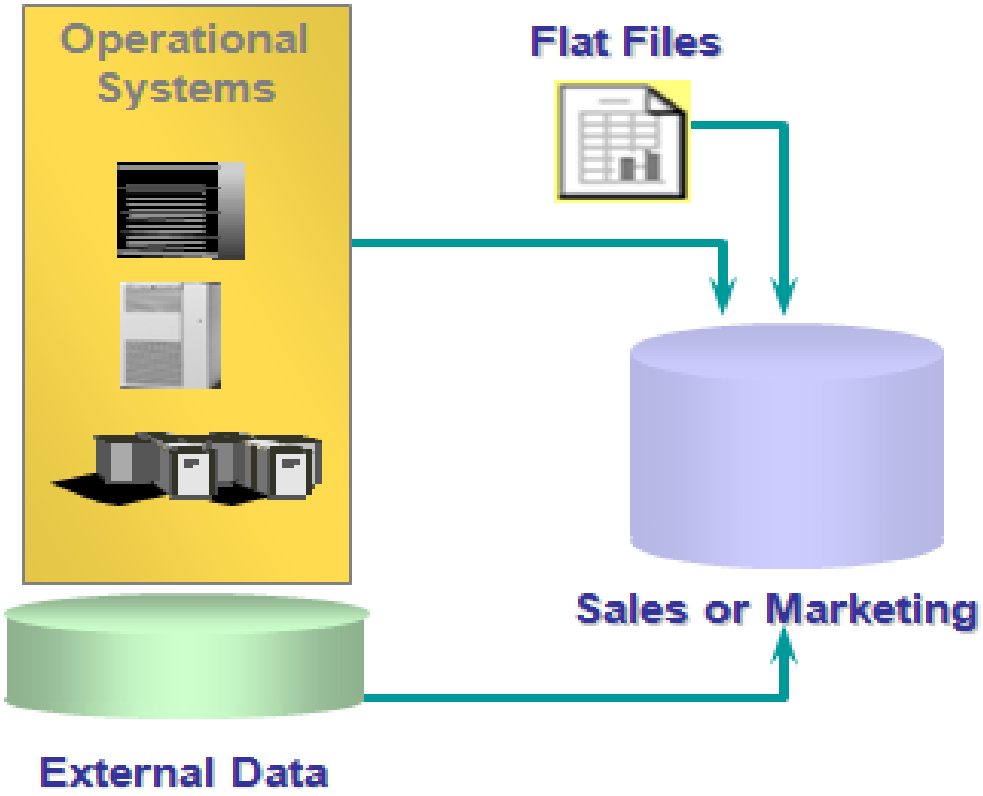
Characteristics

- Do not normally contain detailed operational data unlike data warehouses.
- May contain certain levels of aggregation

Dependent Data Mart



Independent Data Mart



Reasons for Creating a Data Mart

- To give users more flexible access to the data they need to analyse most often.
- To provide data in a form that matches the <u>collective view</u> of a group of users
- To improve end-user response time.
- Potential users of a data mart are clearly defined and can be targeted for support

Reasons for Creating a Data Mart

- To provide appropriately structured data as dictated by the requirements of the end-user access tools.
- Building a data mart is simpler compared with establishing a corporate data warehouse.
- The cost of implementing data marts is far less than that required to establish a data warehouse.

Data Marts Issues

- **Data mart functionality**
- **Data mart size**
- **Data mart load performance**
- **Users access to data in multiple data marts**
- **Data mart Internet / Intranet access**
- **Data mart administration**
- **Data mart installation**

Data Warehouse vs. A Data Mart

Category	Data Warehouse	Data Mart
Scope	Corporate	Line of Business
Subject	Multiple	Single
Data sources	Many	Few
Size	100 GB-TB+	< 100 GB
Implementation Time	Months to Year	Months

Meta Data

What is Meta Data?

- Meta data describes the structure of the contents of a database
- it is the information about the data in the data warehouse. I.e. it is the information of data about data.
- It acts as a quick reference for the data present in a data warehouse.
- Table of content for the data.
- The nerve center i.e. Metadata is placed in a key position and enables communication among various processes

Importance of Meta Data

- Necessary for using, building, and administrating data warehouse.
- Without metadata support, user cannot get an information every time they needed (user create their own ad hoc report).
- Today data warehouse are much larger in size, wide scope so user critically need meta data.
- Metadata provides the means for resource discovery, grouping, filtering, matching user needs

Lifecycle of metadata

- **The lifecycle of metadata** is divided into three phases :
- Collection,
- Maintenance, and
- Equipment.

Collection

- The collection phase's main job is to identify metadata and input the Meta data into central repository.
- The collection of metadata should be done automatically as much as possible so that there is higher reality for collection of metadata.
- The collection of metadata can be done automatic, but some metadata has to be collected manually.

Maintenance

- During maintenance phase metadata must keep track of actual change of data.
- For example if the structure of relational database table changes the metadata should describe the table change and must be updated in order to image change and it must be done to have metadata in good condition in DW and make the data available for analysis purpose.
- It will bring very high level data automatically for physical metadata imaging the structure of data source and data warehouse.

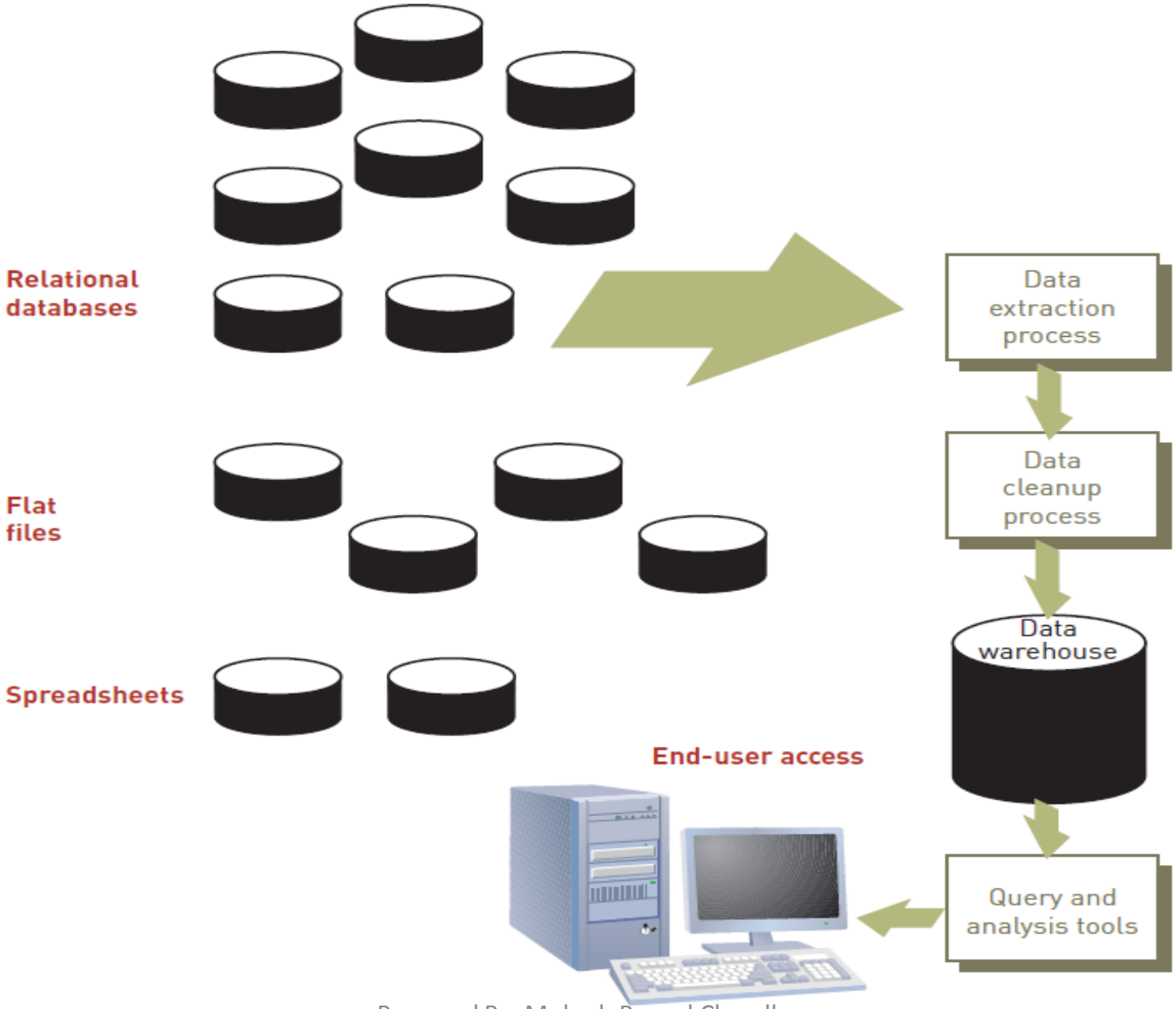
Equipment

- Equipment phase is to provide proper metadata and tools which can be applied on.
- It is the phase to yield after paying out a lot of work in the phases of collection and maintenance metadata.
- The important key factor is to equip metadata to match correct metadata and specifically according to users demand.

Uses of metadata

- By information specialists
  - Describing non-traditional materials
  - Cataloging Web sites
  - Navigating digital objects
  - Managing digital objects over the long term
  - Managing corporate assets
- By novices
  - Preparing Web sites for search engines
  - Describing Eprints
  - iTunes

From Tables and Spreadsheets to Data Cubes



The process of logical design involves arranging data into a series of logical relationships called entities and attributes.

An **entity** represents a chunk of information. In relational databases, an entity often maps to a table.

An **attribute** is a component of an entity that helps define the uniqueness of the entity. In relational databases, an attribute maps to a column.

Relational database model's structural and data independence enables us to view data *logically* rather than *physically*.

- The logical view allows a simpler file concept of data storage.
- The use of logically independent tables is easier to understand.
- Logical simplicity yields simpler and more effective database design methodologies.

An entity is a person, place, event, or thing for which we intend to collect data.

- **University** -- Students, Faculty Members, Courses
- **Airlines** -- Pilots, Aircraft, Routes, Suppliers

Each entity has certain characteristics known as attributes.

- **Student** -- Student Number, Name, GPA, Date of Enrollment, Data of Birth, Home Address, Phone Number, Major
- **Aircraft** -- Aircraft Number, Date of Last Maintenance, Total Hours Flown, Hours Flown since Last Maintenance

A grouping of related entities becomes an entity set.

- The STUDENT entity set contains all student entities.
- The FACULTY entity set contains all faculty entities.
- The AIRCRAFT entity set contains all aircraft entities

A table contains a group of related entities -- i.e. an entity set.

The terms entity set and table are often used interchangeably.

A table is also called a relation.

While entity-relationship diagramming has traditionally been associated with highly normalized models such as OLTP applications, the technique is still useful for data warehouse design in the form of dimensional modeling.

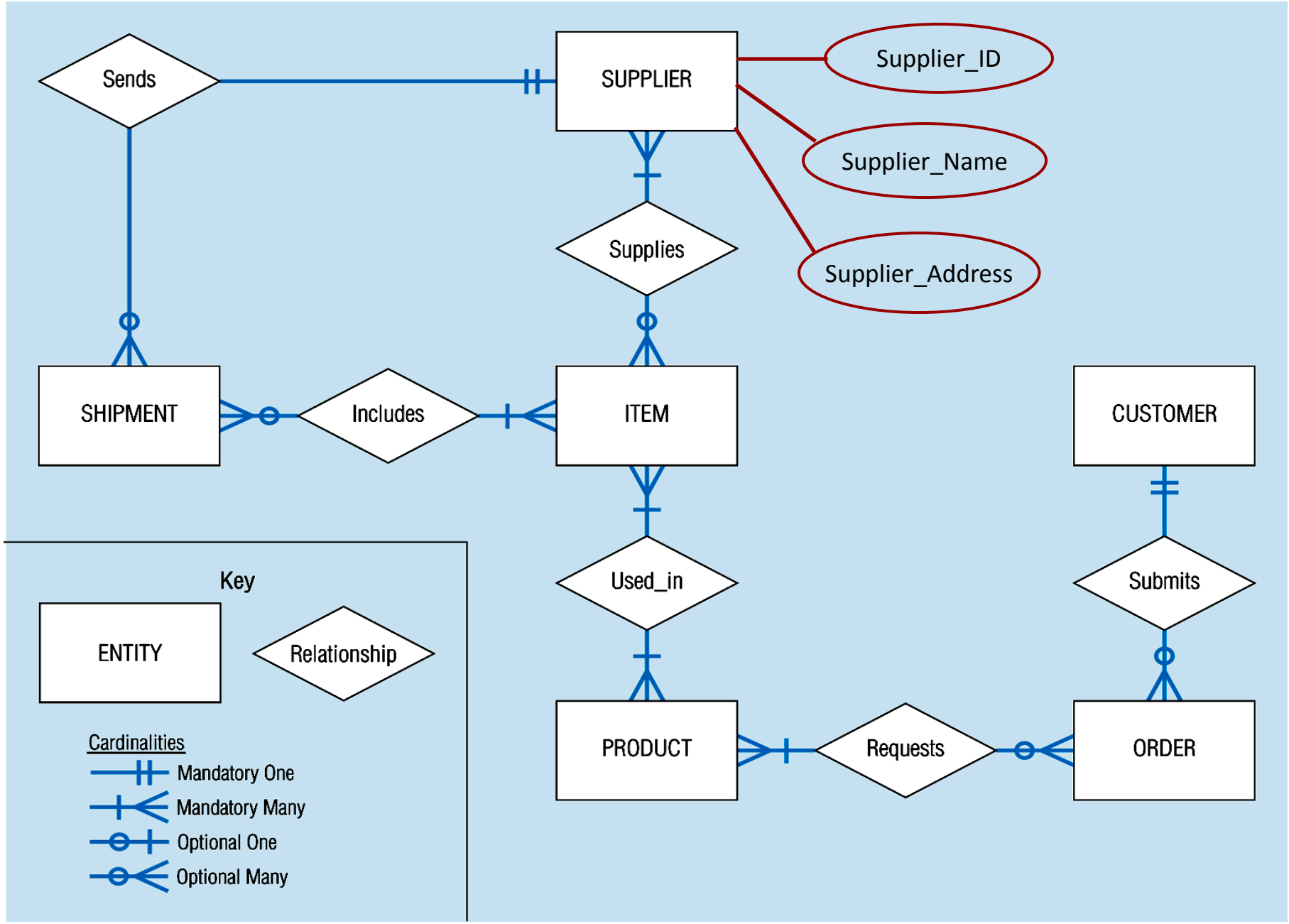


Figure: Sample E-R Diagram

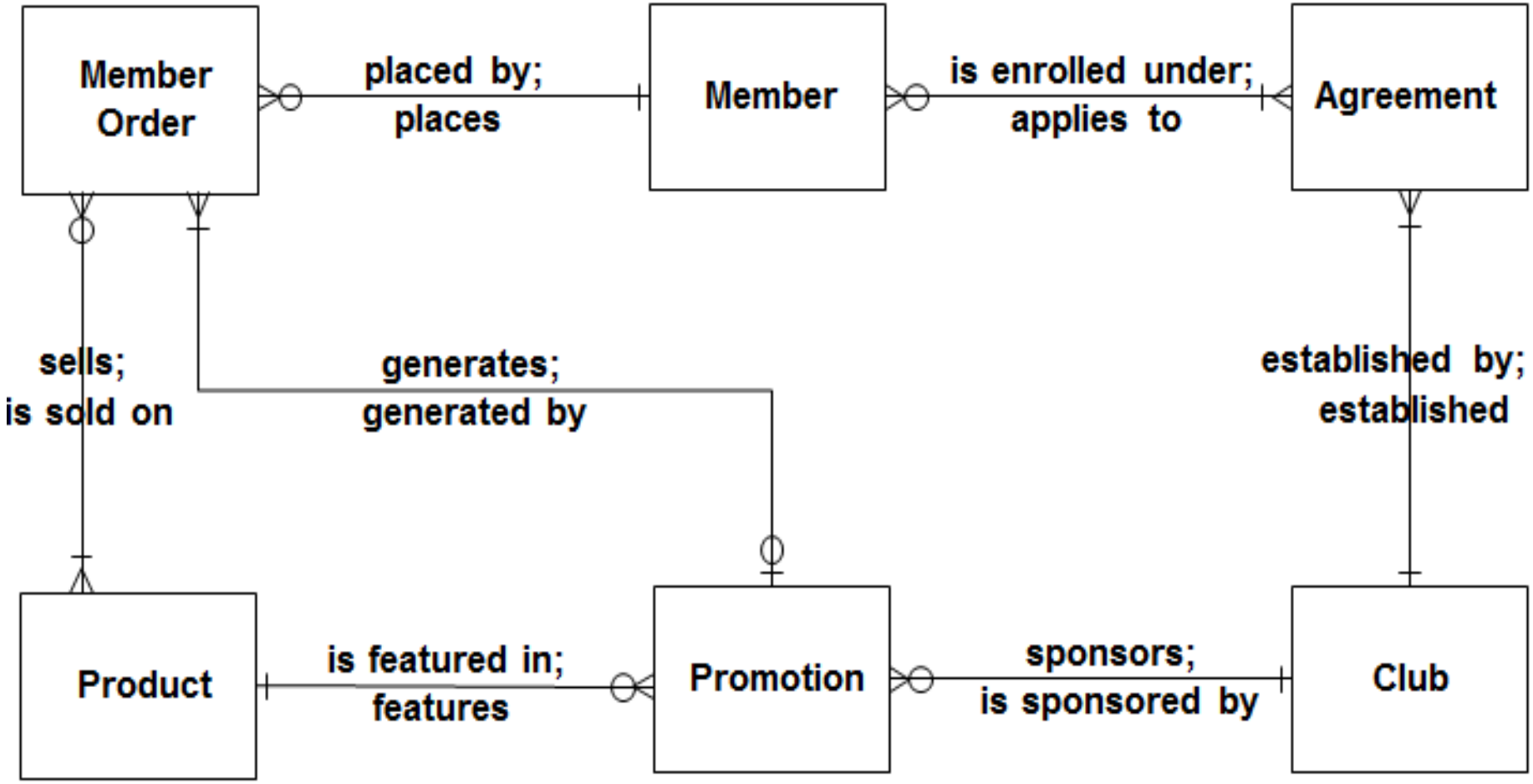
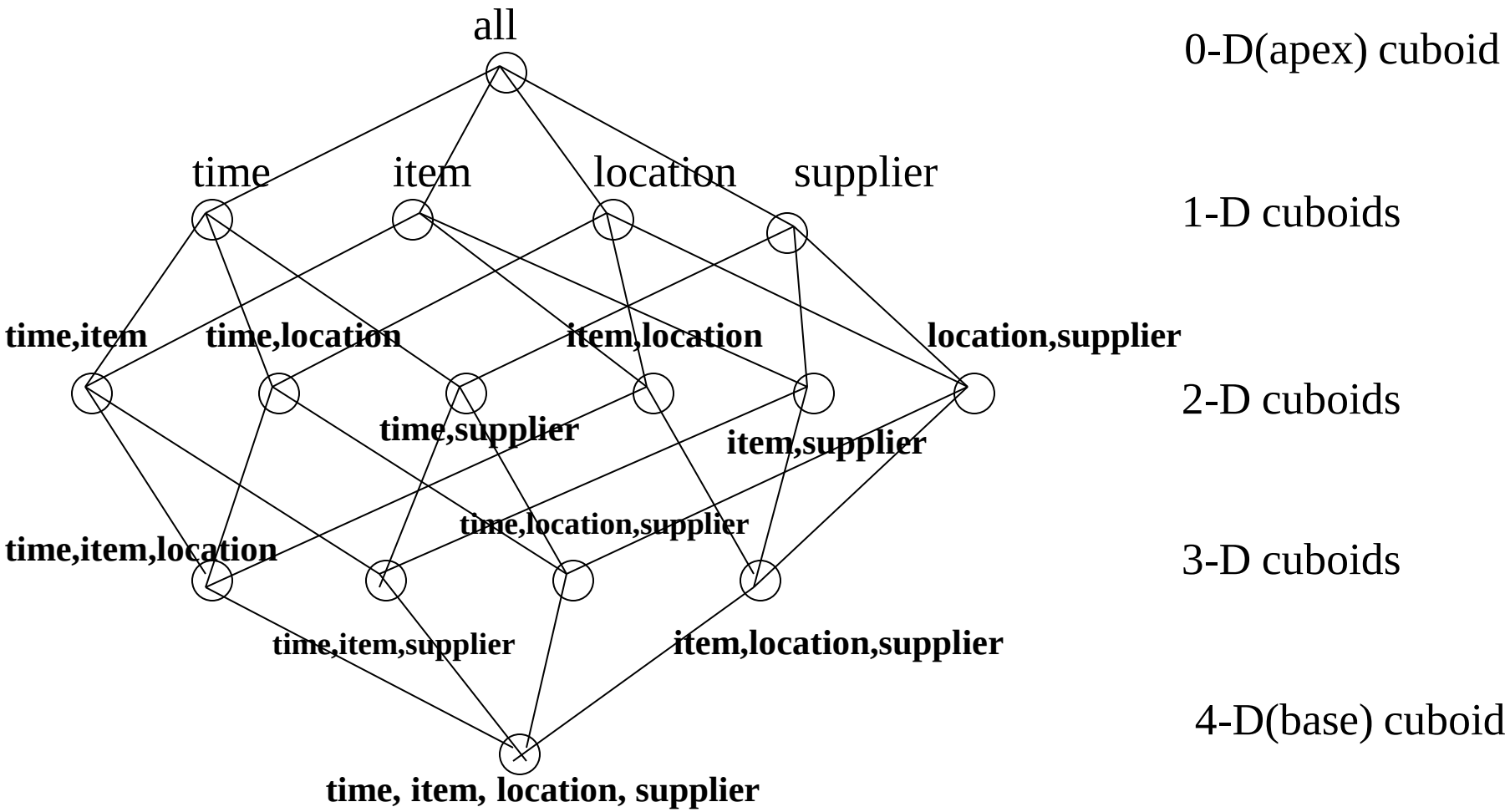


Figure: Sample E-R Diagram

- A data warehouse is based on a multidimensional data model which views data in the form of a data cube
- A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions
  - Dimension tables, such as item (item_name, brand, type), or time(day, week, month, quarter, year)
  - Fact table contains measures (such as dollars_sold) and keys to each of the related dimension tables
- The lattice of cuboids forms a data cube.

Cube: A Lattice of Cuboids



Dimensions and Measures

The database component of a data warehouse is described using a technique called dimensionality modelling.

Every dimensional model (DM) is composed of one table with a composite primary key, called the fact table, and a set of smaller tables called dimension tables.

Each dimension table has a simple (non-composite) primary key that corresponds exactly to one of the components of the composite key in the fact table.

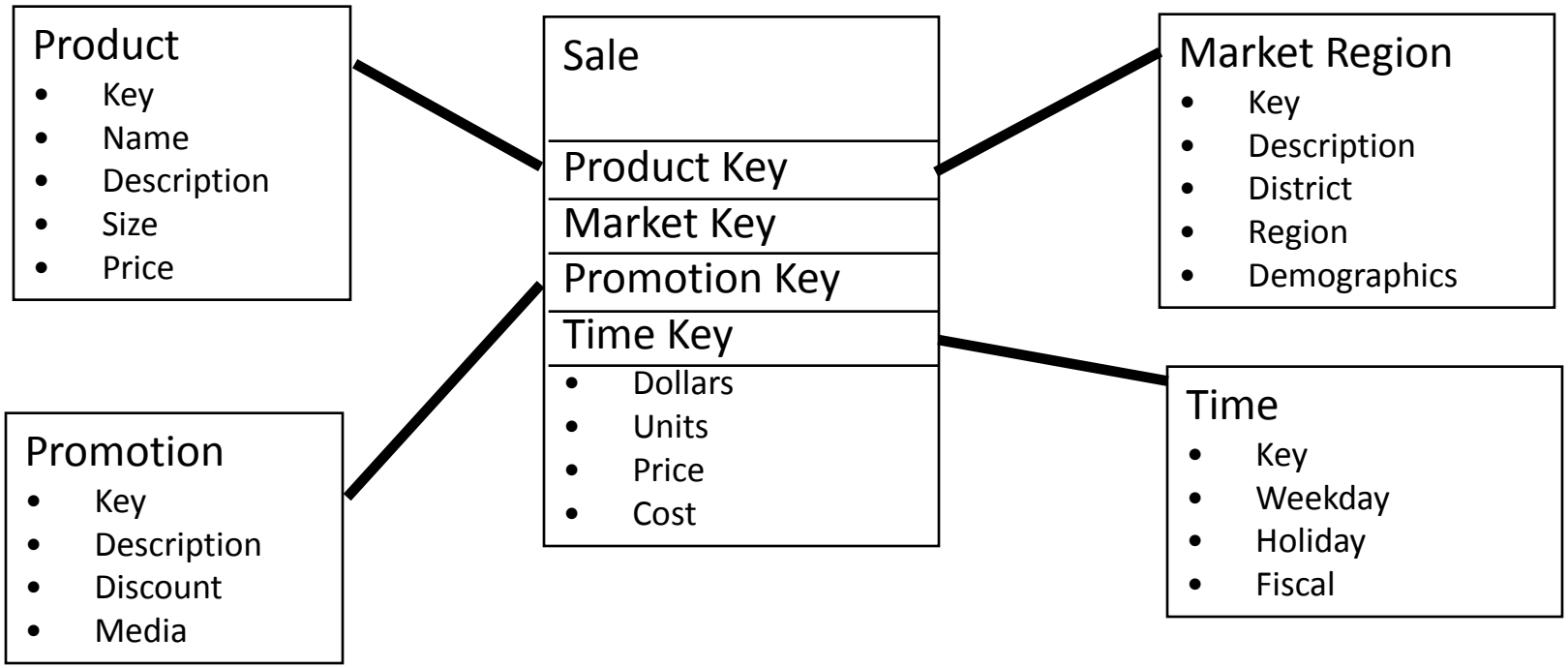
Fact Tables

- A fact table is composed of two or more primary keys and usually also contains numeric data. Because it always contains at least two primary keys it is always a M-M relationship.
- Fact tables contain business event details for summarization.
- Because dimension tables contain records that describe facts, the fact table can be reduced to columns for dimension foreign keys and numeric fact values. Text and de-normalized data are typically not stored in the fact table.

- The logical model for a fact table contains a foreign key column for the primary keys of each dimension.
- The combination of these foreign keys defines the primary key for the fact table.
- Fact tables are often very large, containing hundreds of millions of rows and consuming hundreds of gigabytes or multiple terabytes of storage.

Dimension Tables

- Dimension tables encapsulate the attributes associated with facts and separate these attributes into logically distinct groupings, such as time, geography, products, customers, and so forth.
- A dimension table may be used in multiple places if the data warehouse contains multiple fact tables or contributes data to data marts.
- The data in a dimension is usually hierarchical in nature.
- Hierarchies are determined by the business need to group and summarize data into usable information.
- For example, a time dimension often contains the hierarchy elements: (all time), Year, Quarter, Month, Day, or (all time), Year Quarter, Week, Day.



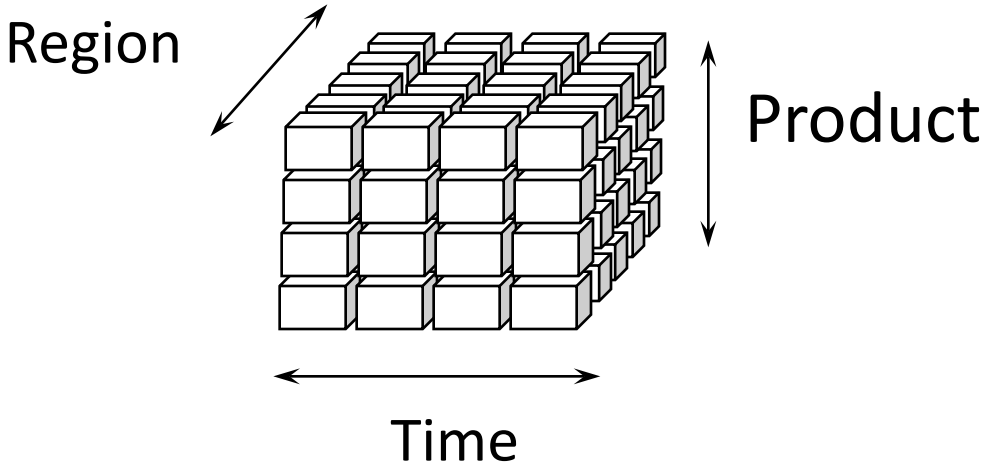


Figure: Dimensional Model

A Data Mining Query Language, DMQL: Language Primitives

Cube Definition (Fact Table)

define cube <cube_name> [<dimension_list>]: <measure_list>

Dimension Definition (Dimension Table)

define dimension <dimension_name> as
(<attribute_or_subdimension_list>)

Special Case (Shared Dimension Tables)

- First time as "cube definition"
- define dimension <dimension_name> as <dimension_name_first_time> in cube <cube_name_first_time>

Data Warehouse Schema

- A data warehouse, however, requires a concise, subject-oriented schema that facilitates on-line data processing (OLAP).
- The most popular data model for a data warehouse is **a multidimensional model.**
- Such a model can exist in the following forms
  - a star schema
  - a snowflake schema
  - a fact constellation schema.
- The major focus will be on the star schema which is commonly used in the design of many data warehouse.

Star Schema

- The star schema is a data modeling technique used to map multidimensional decision support into a relational database.
- Star schemas yield an easily implemented model for multidimensional data analysis while still preserving the relational structure of the operational database.
- Others name: star-join schema, data cube, data list, grid file and multi-dimension schema

Excellent for ad-hoc queries, but bad for online transaction processing

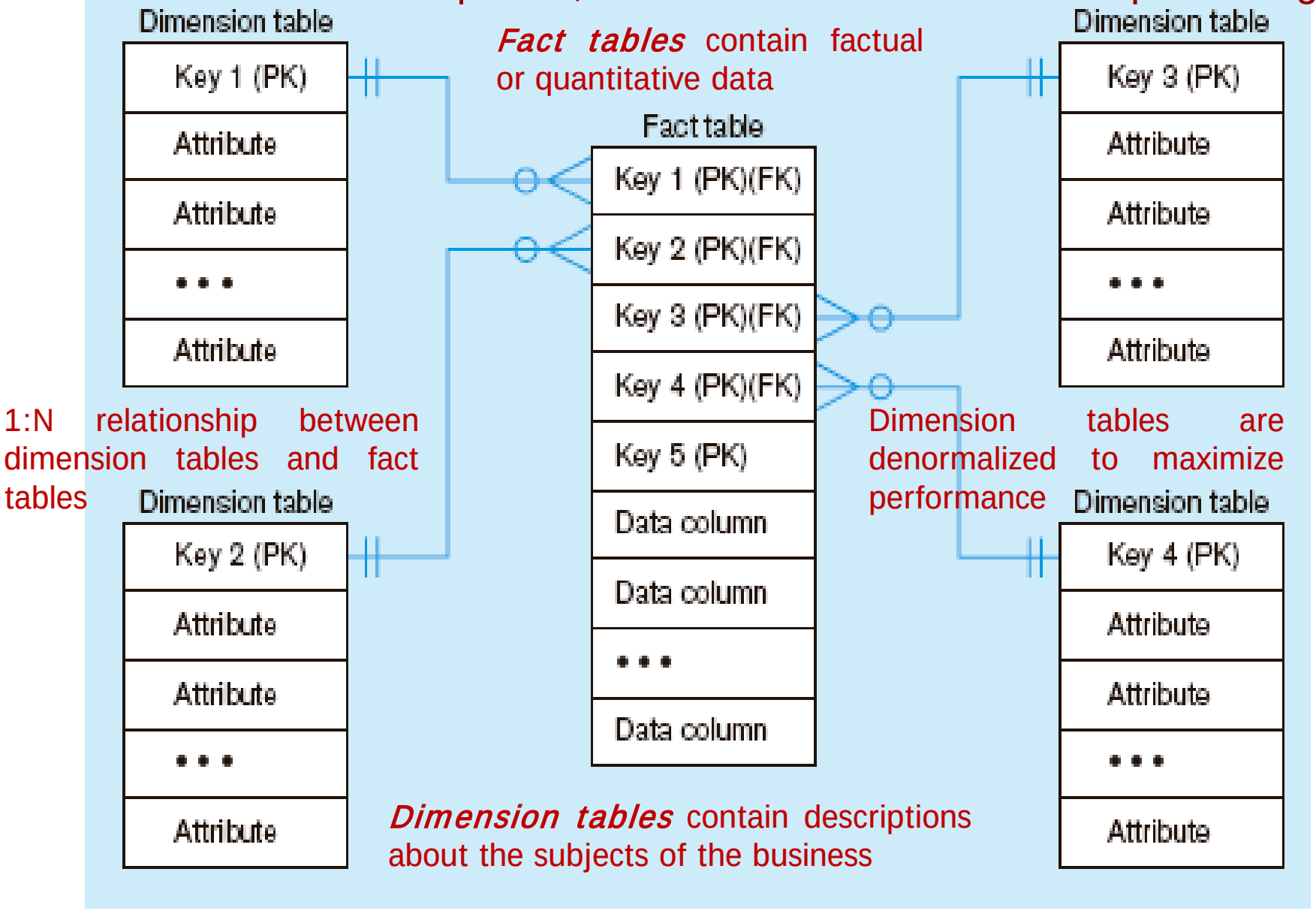


Figure: Components of a **star schema**

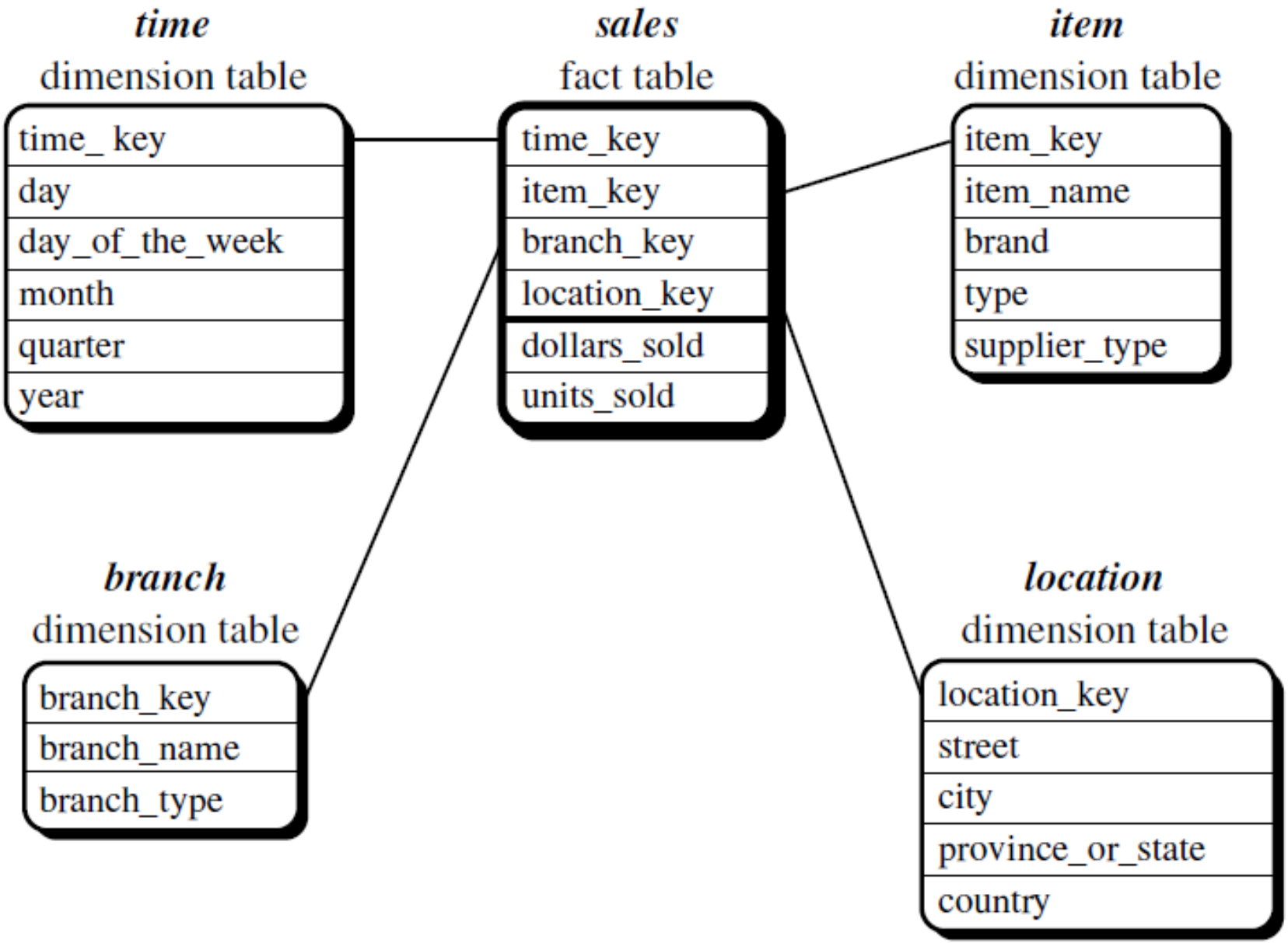


Figure: Star schema of a data warehouse for sales

- The schema contains a central fact table for *sales* that contains keys to each of the four dimensions, along with two measures: *dollars_sold, avg_sales,* and *units_sold*.
- To minimize the size of the fact table, dimension identifiers (such as *time key* and *item key*) are system-generated identifiers.
- Notice that in the star schema, each dimension is represented by only one table, and each table contains a set of attributes.
- For example, the *location* dimension table contains the attribute set {*location key, street, city, province or state, country*}

Defining a Star Schema in DMQL

define cube sales_star [time, item, branch, location]:

dollars_sold = sum(sales_in_dollars), avg_sales = avg(sales_in_dollars), units_sold = count(*)

define dimension time as (time_key, day, day_of_week, month, quarter, year)

define dimension item as (item_key, item_name, brand, type, supplier_type)

define dimension branch as (branch_key, branch_name, branch_type)

define dimension location as (location_key, street, city, province_or_state, country)

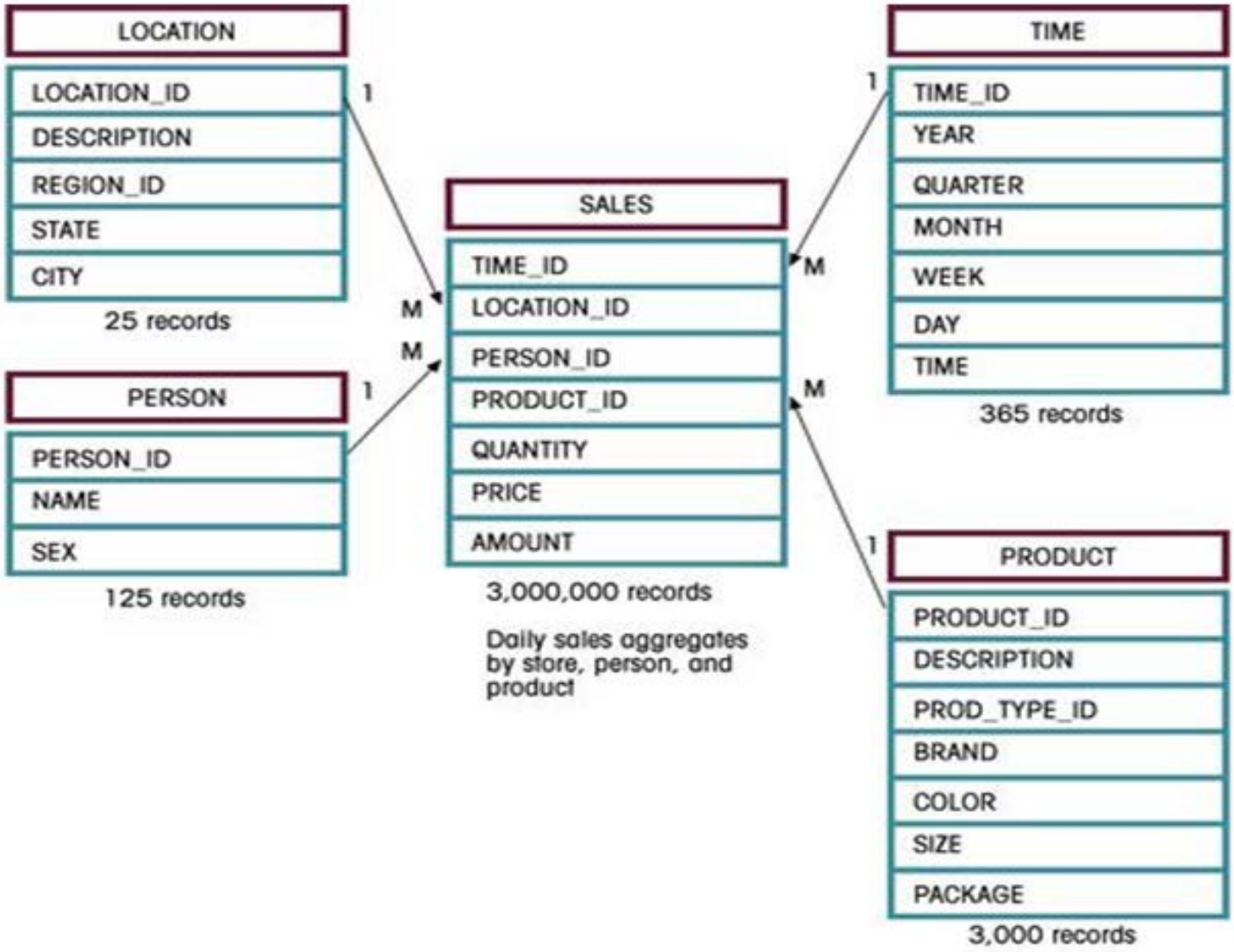


Figure: Star Schema for Sales

Advantages of Star Schema

- Star Schema is very easy to understand, even for non technical business manager.
- Star Schema provides better performance and smaller query times
- Star Schema is easily extensible and will handle future changes easily

Issues Regarding Star Schema

- Dimension table keys must be ***surrogate*** (non-intelligent and non-business related), because:
  - Keys may change over time
  - Length/format consistency
- Granularity of Fact Table–what level of detail do you want?
  - Transactional grain–finest level
  - Aggregated grain–more summarized
  - Finer grains ➔ better ***market basket analysis*** capability
  - Finer grain ➔ more dimension tables, more rows in fact table
- Duration of the database–how much history should be kept?
  - Natural duration–13 months or 5 quarters
  - Financial institutions may need longer duration
  - Older data is more difficult to source and cleanse

Snowflake Schema

A schema is called a *snowflake schema* if one or more dimension tables do not join directly to the fact table but must join through other dimension tables.

It is a variant of star schema model. It has a single, large and central fact table and one or more tables for each dimension.

Characteristics:

- Normalization of dimension tables
- Each hierarchical level has its own table
- less memory space is required
- a lot of joins can be required if they involve attributes in secondary dimension tables

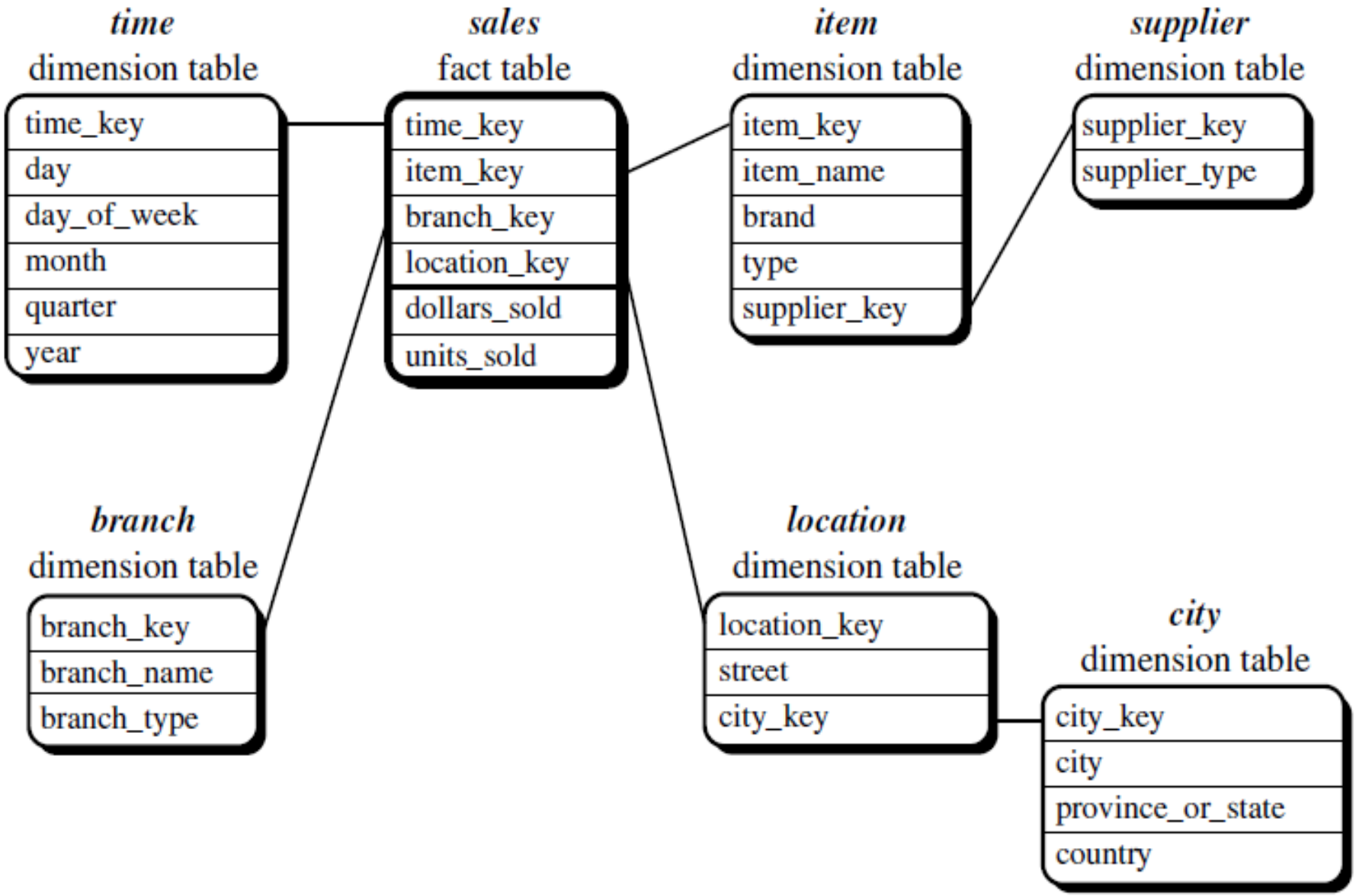


Figure: Snowflake schema of a data warehouse for sales

Defining a Snowflake Schema in DMQL

define cube sales_snowflake [time, item, branch, location]:

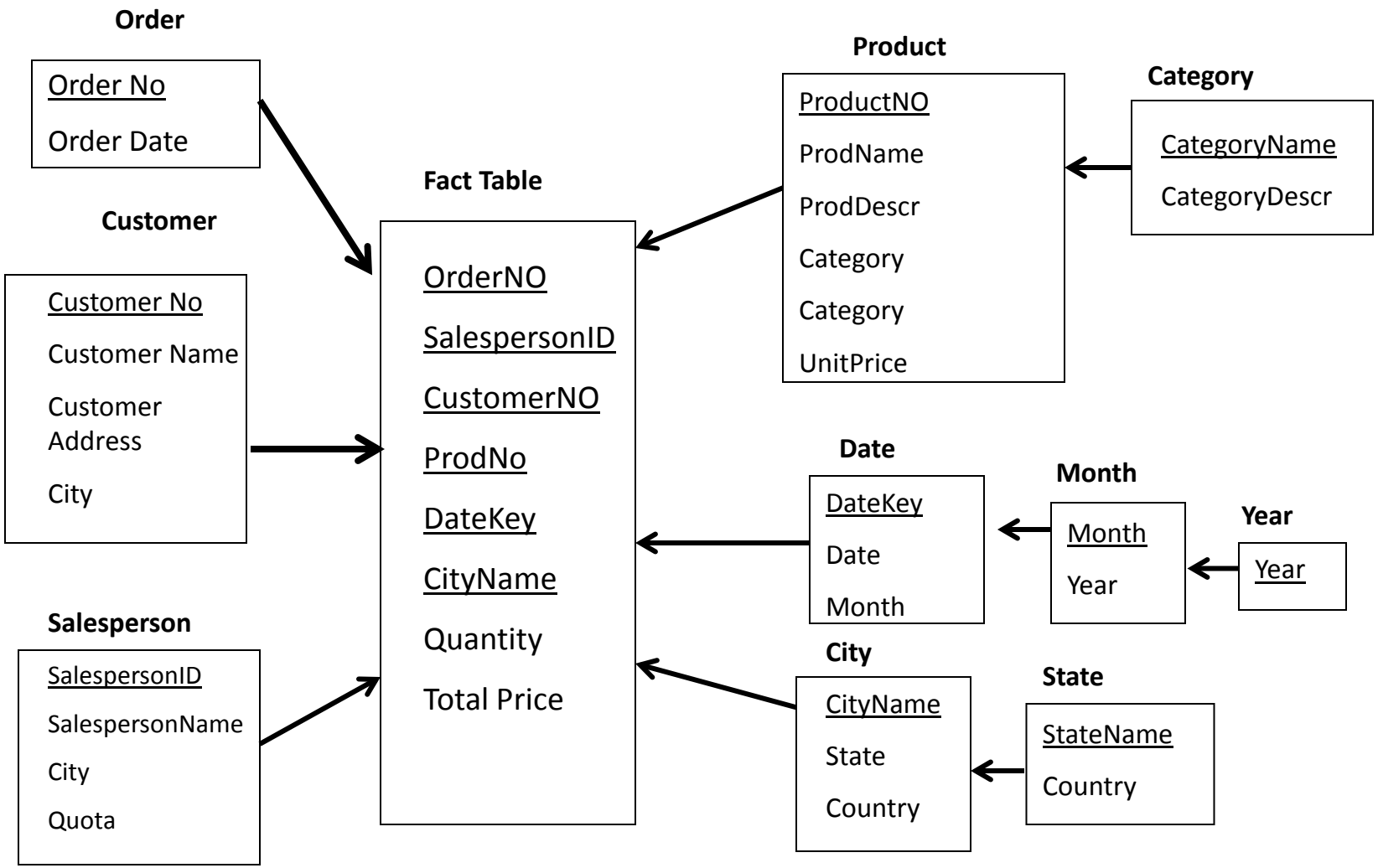
dollars_sold = sum(sales_in_dollars), avg_sales = avg(sales_in_dollars), units_sold = count(*)

define dimension time as (time_key, day, day_of_week, month, quarter, year)

define dimension item as (item_key, item_name, brand, type, supplier(supplier_key, supplier_type))

define dimension branch as (branch_key, branch_name, branch_type)

define dimension location as (location_key, street, city(city_key, province_or_state, country))


Order
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Order Date
Customer
Customer No
Customer Name
Customer Address
City
Salesperson
SalespersonID
SalespersonName
City
Quota
Fact Table
OrderNO
SalespersonID
CustomerNO
ProdNo
DateKey
CityName
Quantity
Total Price
Product
ProductNO
ProdName
ProdDescr
Category
Category
UnitPrice
Category
CategoryName
CategoryDescr
Date
DateKey
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Year
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City
CityName
State
Country
State
StateName
Country

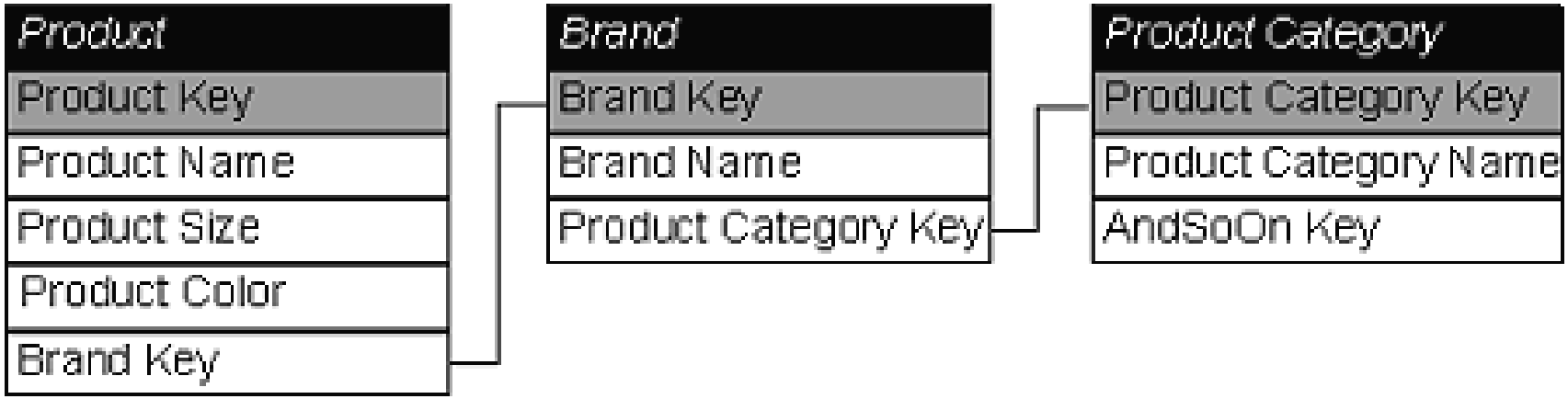


Figure: Snowflake Schema

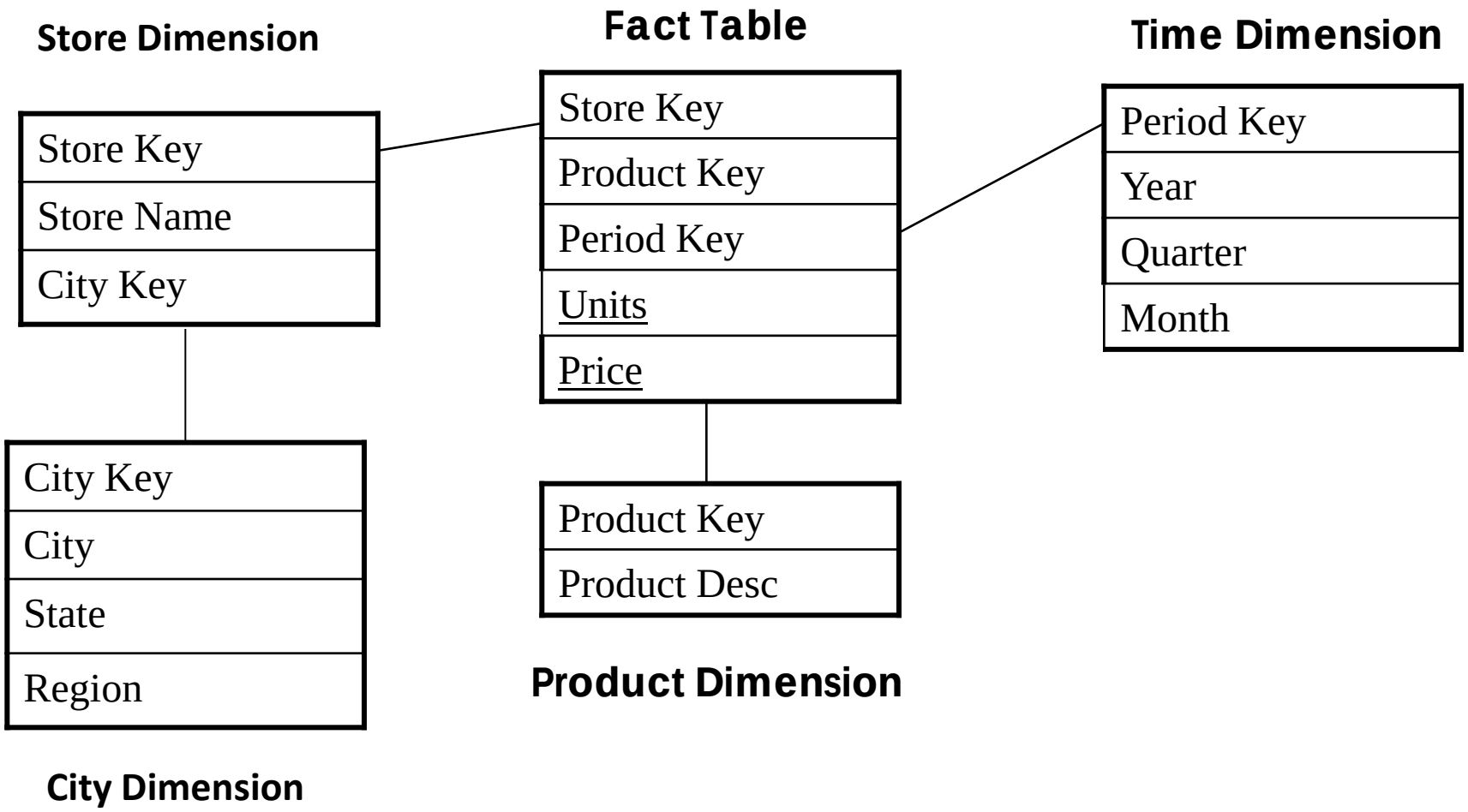


Figure: Snowflake Schema

Difference between Star Schema and Snow-flake Schema

- Star Schema is a multi-dimension model where each of its disjoint dimension is represented in single table.
- Snow-flake is normalized multi-dimension schema when each of disjoint dimension is represent in multiple tables.
- Star schema can become a snow-flake
- Both star and snowflake schemas are dimensional models; the difference is in their physical implementations.
- Snowflake schemas support ease of dimension maintenance because they are more normalized.
- Star schemas are easier for direct user access and often support simpler and more efficient queries.
- It may be better to create a star version of the snowflaked dimension for presentation to the users

Fact-Constellation Schema

- Multiple fact tables share dimension tables.
- This schema is viewed as collection of stars hence called as galaxy schema or fact constellation.
- Sophisticated application requires such schema.
- In the Fact Constellations, aggregate tables are created separately from the detail, therefore, it is impossible to pick up, for example, Store detail when querying the District Fact Table.
- Fact Constellation is a good alternative to the Star, but when dimensions have very high cardinality, the sub-selects in the dimension tables can be a source of delay.

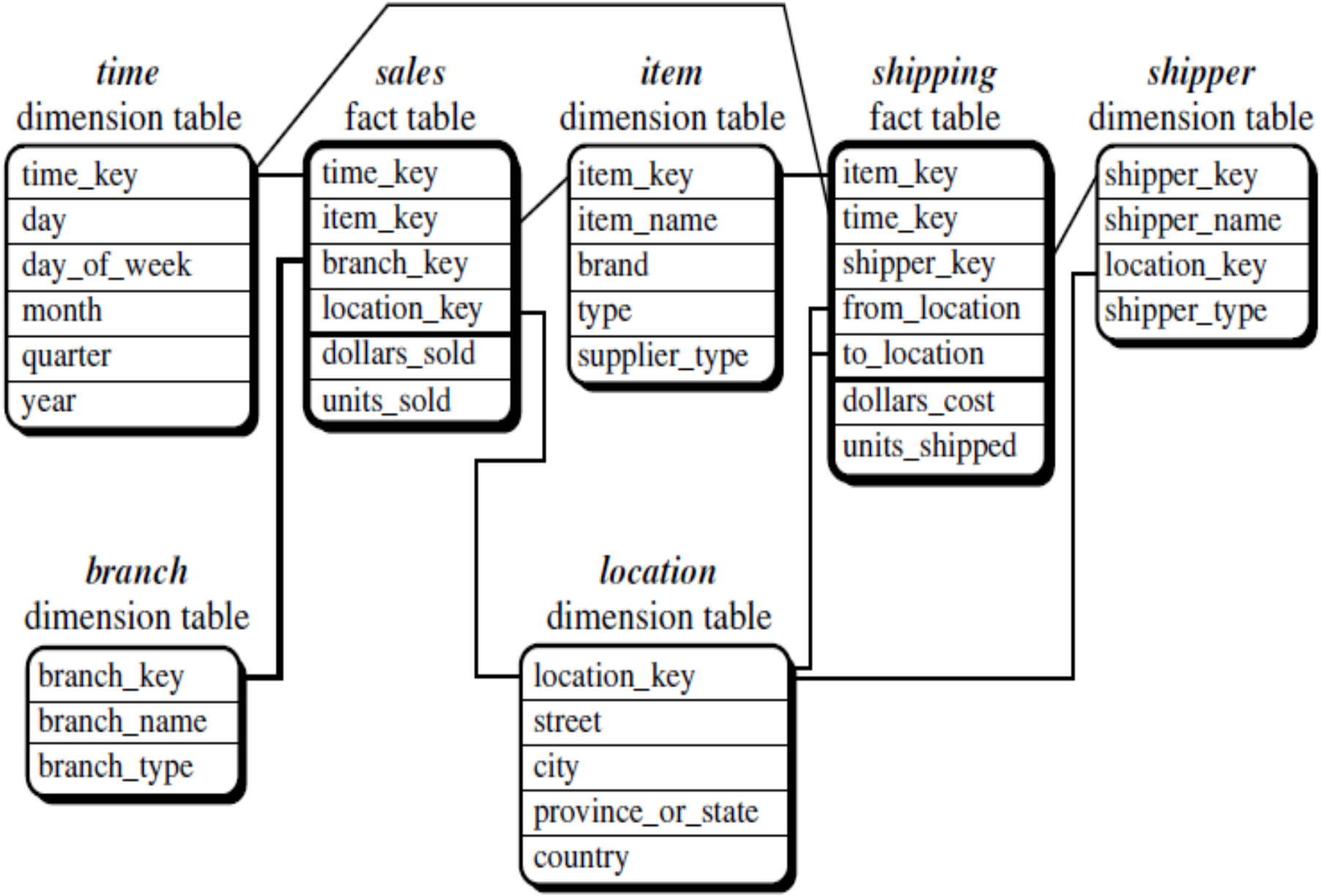
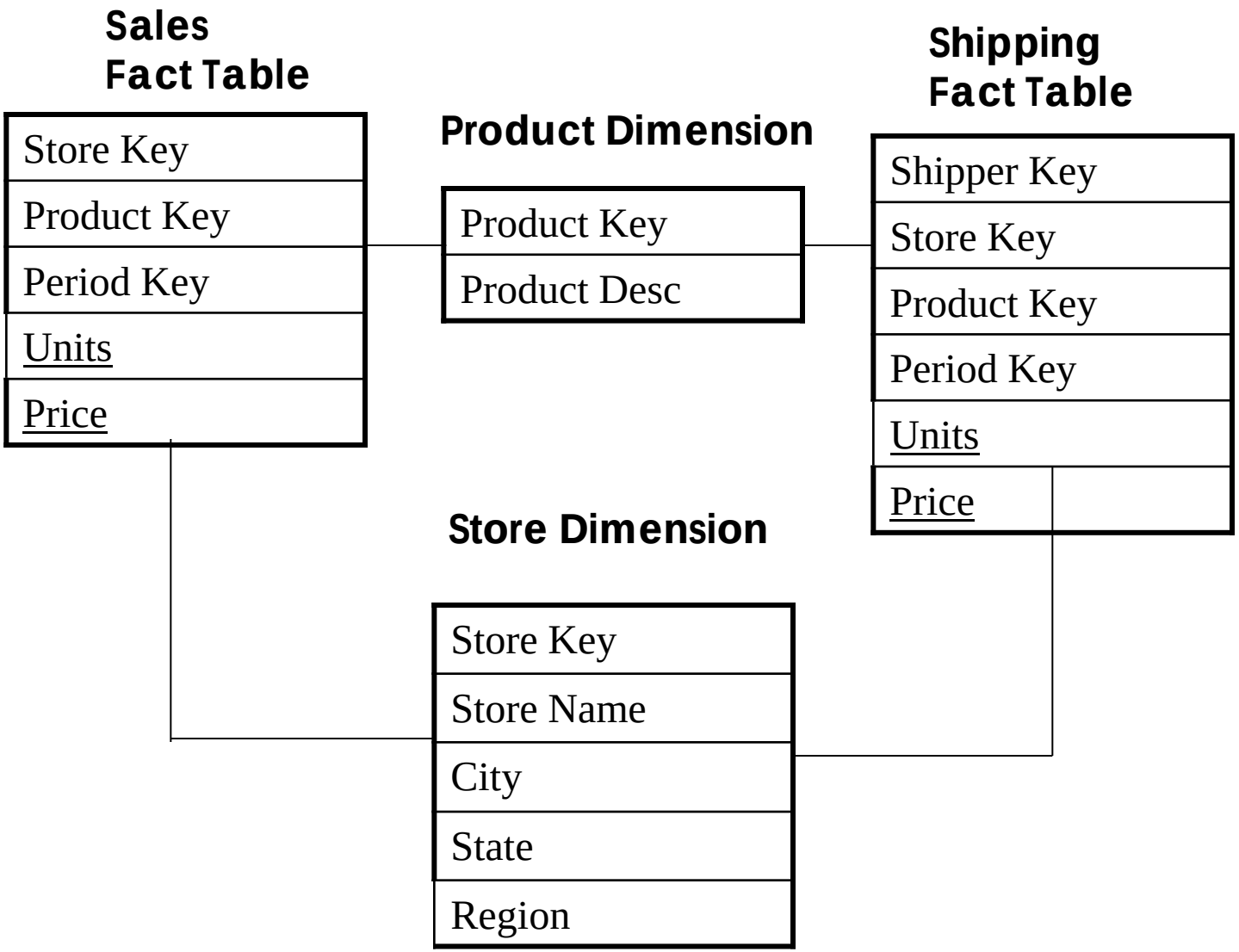


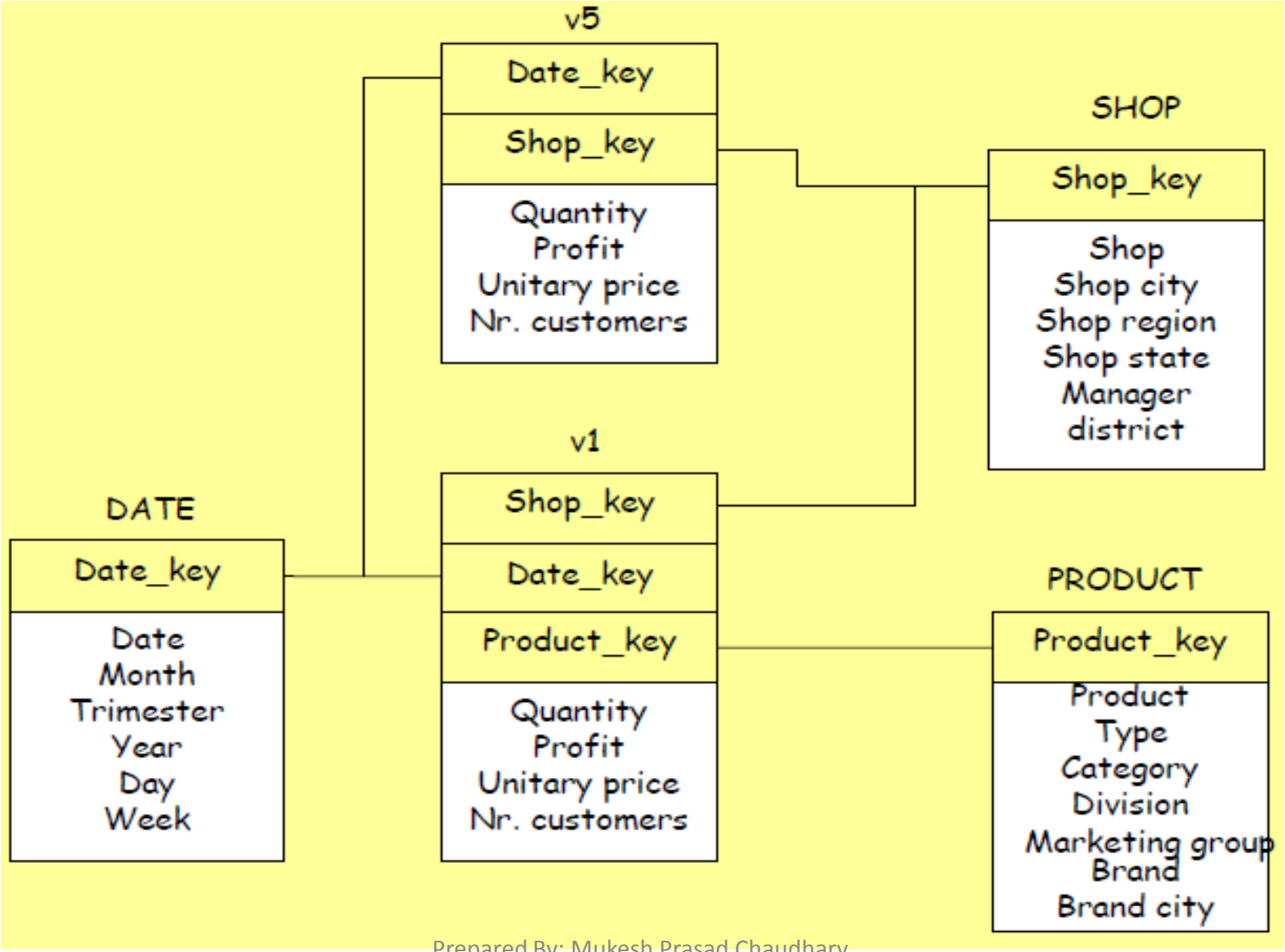
Figure: Fact constellation schema of a data warehouse for sales and shipping

Defining a Fact Constellation in DMQL

```
define cube sales [time, item, branch, location]:
        dollars_sold = sum(sales_in_dollars), avg_sales = avg(sales_in_dollars),
          units_sold = count(*)
define dimension time as (time_key, day, day_of_week, month, quarter, year)
define dimension item as (item_key, item_name, brand, type, supplier_type)
define dimension branch as (branch_key, branch_name, branch_type)
define dimension location as (location_key, street, city, province_or_state,
    country)
define cube shipping [time, item, shipper, from_location, to_location]:
        dollar_cost = sum(cost_in_dollars), unit_shipped = count(*)
define dimension time as time in cube sales
define dimension item as item in cube sales
define dimension shipper as (shipper_key, shipper_name, location as location in
    cube sales, shipper_type)
define dimension from_location as location in cube sales
define dimension to_location as location in cube sales
```

More Examples on Fact-Constellation Schema




v5
Date_key
Shop_key
Quantity
Profit
Unitary price
Nr. customers
SHOP
Shop_key
Shop
Shop city
Shop region
Shop state
Manager
district
v1
Shop_key
Date_key
Product_key
Quantity
Profit
Unitary price
Nr. customers
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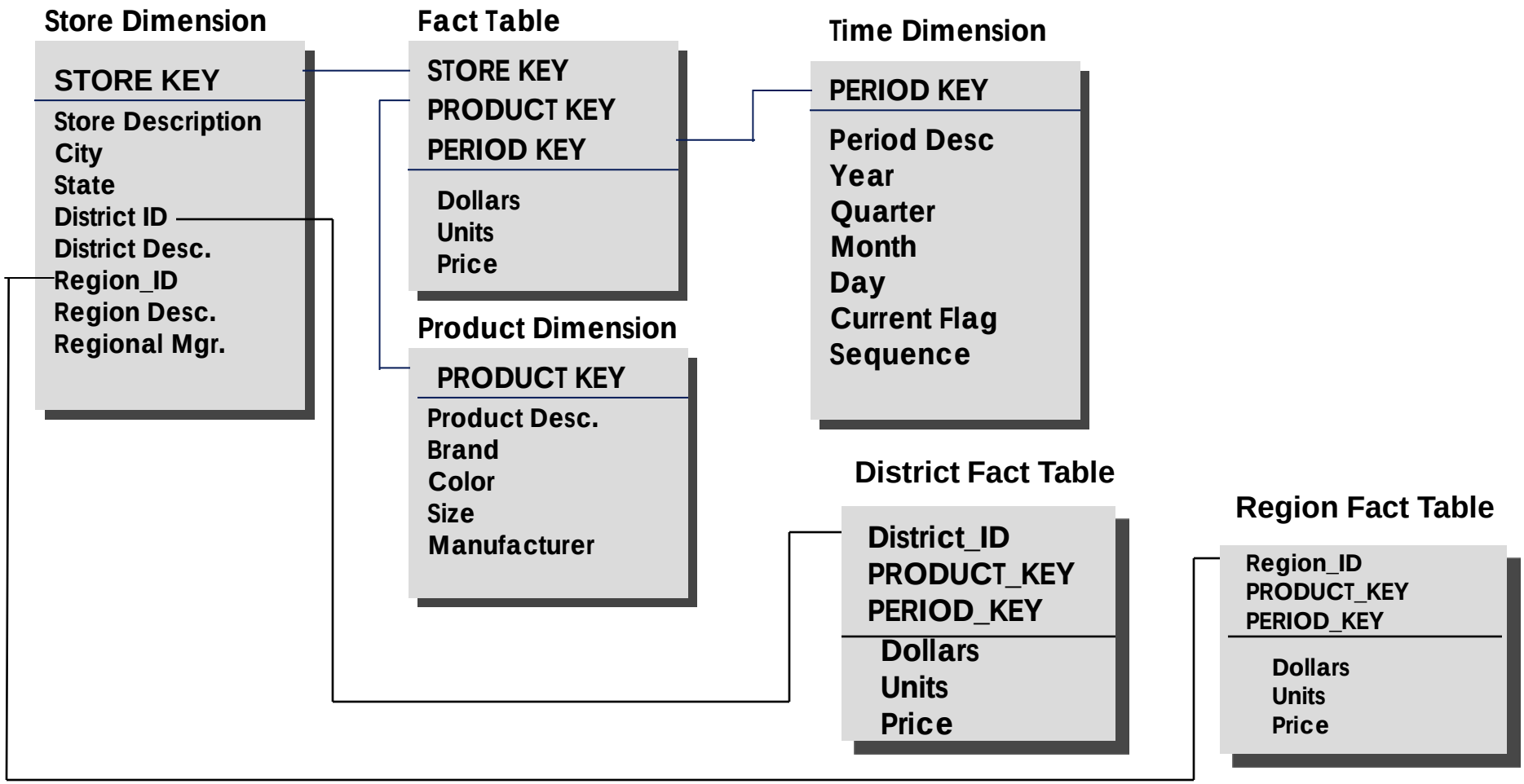
Product_key
Product
Type
Category
Division
Marketing group
Brand
Brand city


Store Dimension
STORE KEY
Store Description
City
State
District ID
District Desc.
Region_ID
Region Desc.
Regional Mgr.
Fact Table
STORE KEY
PRODUCT KEY
PERIOD KEY
Dollars
Units
Price
Time Dimension
PERIOD KEY
Period Desc
Year
Quarter
Month
Day
Current Flag
Sequence
Product Dimension
PRODUCT KEY
Product Desc.
Brand
Color
Size
Manufacturer
District Fact Table
District_ID
PRODUCT_KEY
PERIOD_KEY
Dollars
Units
Price
Region Fact Table
Region_ID
PRODUCT_KEY
PERIOD_KEY
Dollars
Units
Price

Multidimensional Data Model

Data warehouses and OLAP tools are based on a multidimensional data model. This model views data in the form of a *data cube*.

"What is a data cube?" A data cube allows data to be modeled and viewed in multiple dimensions. It is defined by dimensions and facts.

In general terms, dimensions are the perspectives or entities with respect to which an organization wants to keep records.

Each dimension may have a table associated with it, called a dimension table, which further describes the dimension.

A multidimensional data model is typically organized around a central theme, like *sales*, for instance. This theme is represented by a fact table. Facts are numerical measures.

	location = "Vancouver"			
	item (type)			
time (quarter)	home entertainment	computer	phone	security
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580

Table: A 2-D view of sales data according to the dimensions *time* and *item*, where the sales are from branches located in the city of Vancouver. The measure displayed is *dollars sold* (in thousands).

	location = "Chicago"				location = "New York"				location = "Toronto"				location = "Vancouver"			
	item				item				item				item			
	home				home				home				home			
time	ent.	comp.	phone	sec.	ent.	comp.	phone	sec.	ent.	comp.	phone	sec.	ent.	comp.	phone	sec.
Q1	854	882	89	623	1087	968	38	872	818	746	43	591	605	825	14	400
Q2	943	890	64	698	1130	1024	41	925	894	769	52	682	680	952	31	512
Q3	1032	924	59	789	1034	1048	45	1002	940	795	58	728	812	1023	30	501
Q4	1129	992	63	870	1142	1091	54	984	978	864	59	784	927	1038	38	580

Table: A 3-D view of sales data according to the dimensions *time*, *item*, and *location*. The measure displayed is *dollars sold* (in thousands).

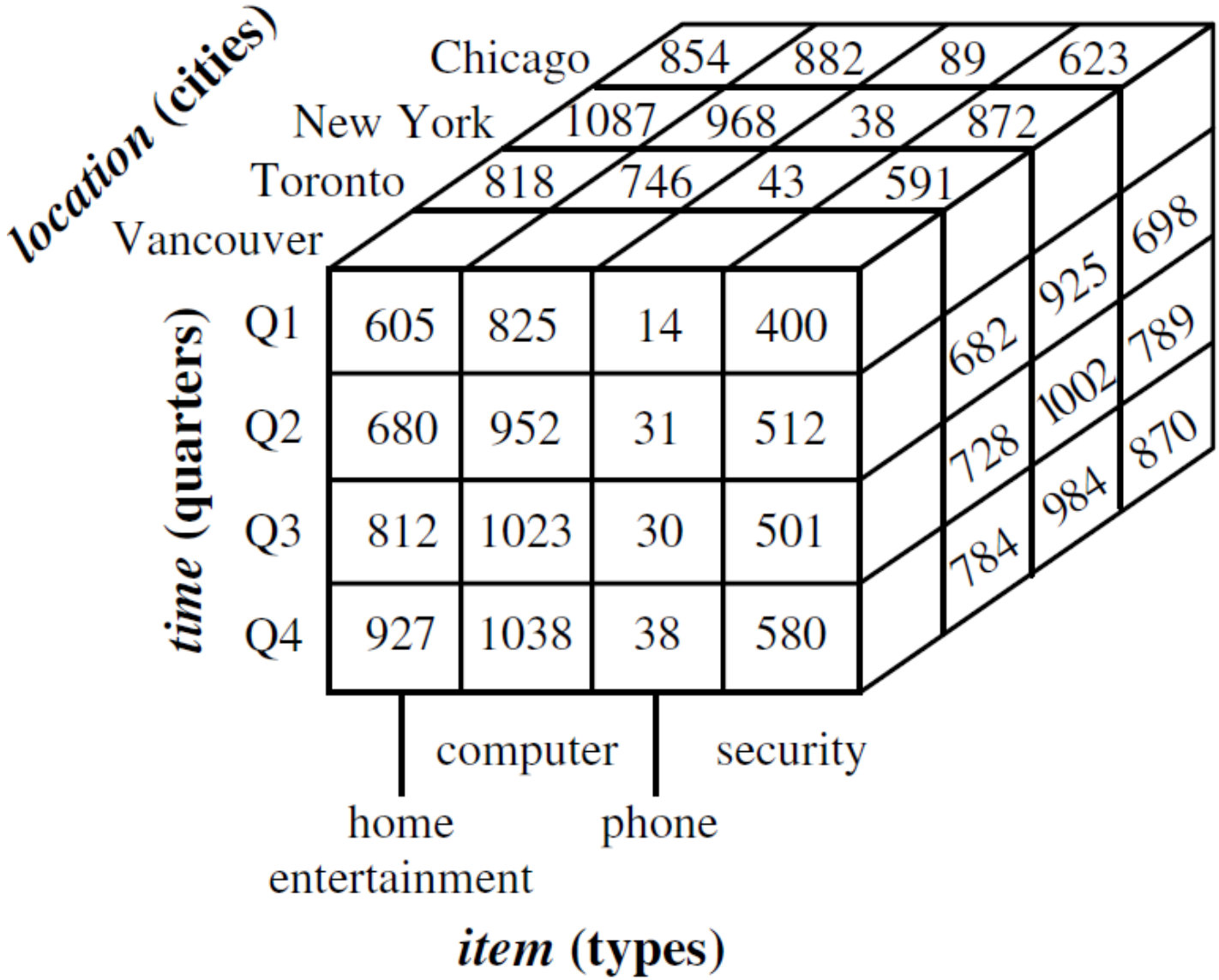


Figure: A 3-D data cube representation of the data in the table above, according to the dimensions *time*, *item*, and *location*. The measure displayed is *dollars sold* (in thousands).

Question?

Suppose that we would now like to view our sales data with an additional fourth dimension, such as *supplier*.

What should we do??

Any Solution???

Solution!!

Viewing things in 4-D becomes tricky. However, we can think of a 4-D cube as being a series of 3-D cubes as shown below:

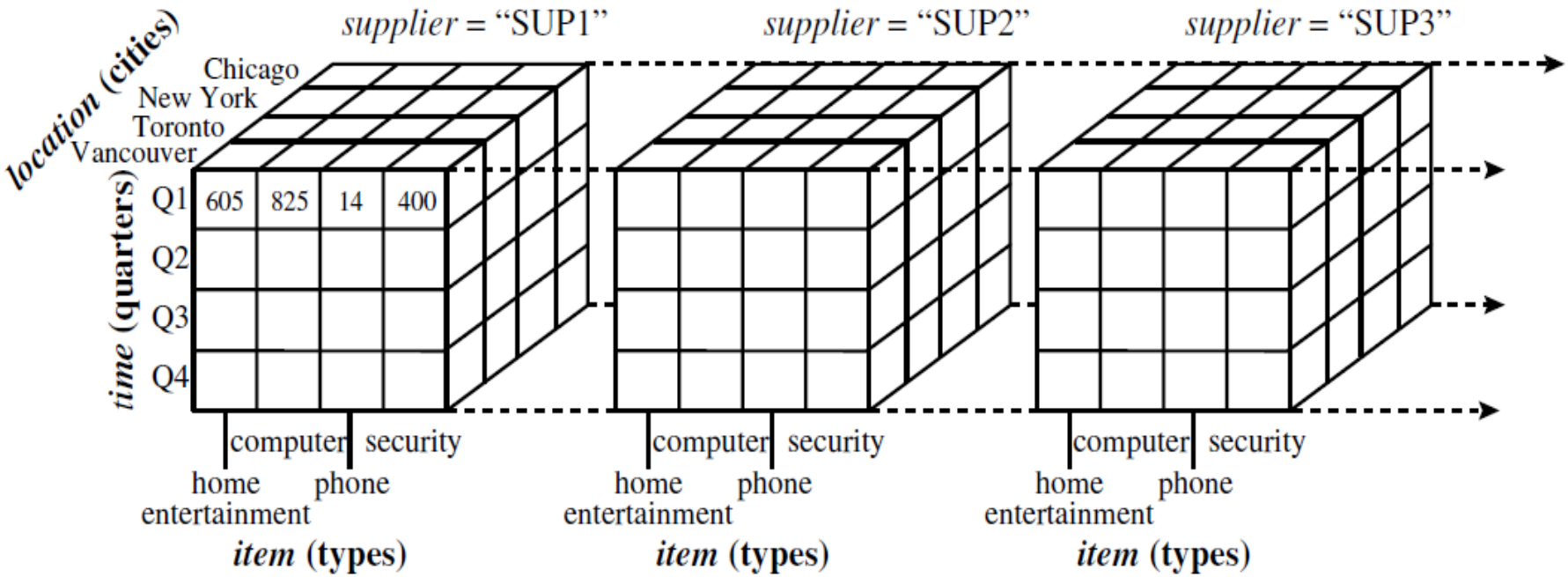


Figure: A 4-D data cube representation of sales data, according to the dimensions *time*, *item*, *location*, and *supplier*. The measure displayed is *dollars sold* (in thousands). For improved readability, only some of the cube values are shown.

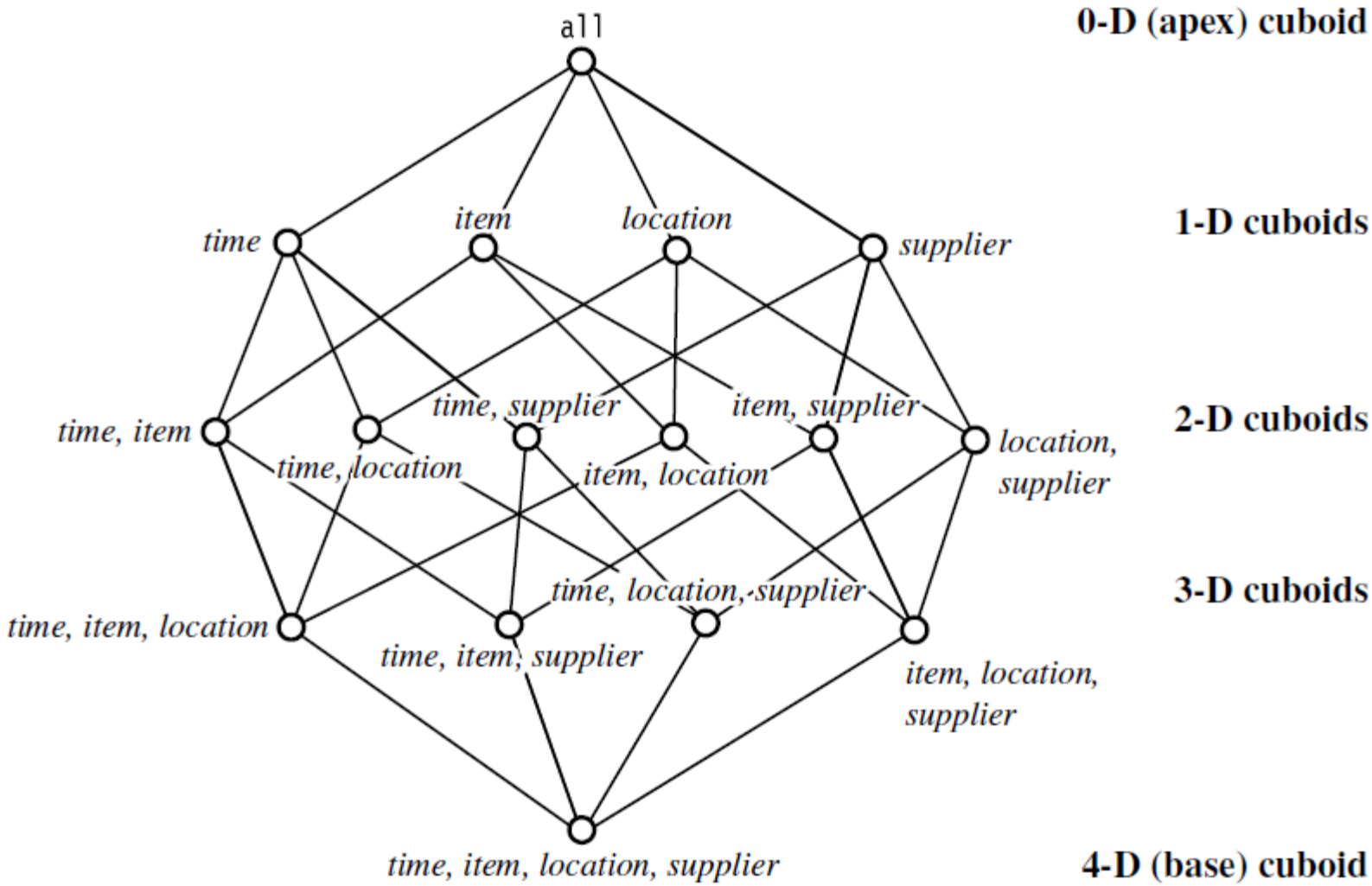


Figure: Lattice of cuboids, making up a 4-D data cube for the dimensions *time*, *item*, *location*, and *supplier*. Each cuboid represents a different degree of summarization.

Class Assignment (1)

Suppose that a data warehouse consists of the three dimensions *time, doctor*, and *patient*, and the two measures *count* and *charge*, where *charge* is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

(b) Draw a schema diagram for the above data warehouse using one of the schema classes listed in (a).

Class Assignment (2)

Suppose that a data warehouse for *XYZ University* consists of the following four dimensions: *student, course, semester*, and *instructor*, and two measures *count* and *avg grade*.

When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the *avg grade* measure stores the actual course grade of the student. At higher conceptual levels, *avg grade* stores the average grade for the given combination.

(a) Draw a *snowflake schema* diagram for the data warehouse.

Class Assignment (3)

A data warehouse can be modeled by either a *star schema* or a *snowflake schema*. Briefly describe the similarities and the differences of the two models, and then analyze their advantages and disadvantages with regard to one another.

Give your opinion of which might be more empirically useful and state the reasons behind your answer.

Class Assignment (4)

Let us consider the case of a real estate agency whose database is composed by the following tables:

OWNER (IDOwner, Name, Surname, Address, City, Phone)

ESTATE (IDEstate, IDOwner, Category, Area, City, Province, Rooms, Bedrooms, Garage, Meters)

CUSTOMER (IDCust, Name, Surname, Budget, Address, City, Phone)

AGENT (IDAgent, Name, Surname, Office, Address, City, Phone)

AGENDA (IDAgent, Data, Hour, IDEstate, ClientName)

VISIT (IDEstate, IDAgent, IDCust, Date, Duration)

SALE (IDEstate, IDAgent, IDCust, Date, AgreedPrice, Status)

RENT (IDEstate, IDAgent, IDCust, Date, Price, Status, Time)

Design a Star Schema or Snowflake Schema for the DW.

Hints:

The following ideas will be used during the solution of the exercise:

- supervisors should be able to control the sales of the agency

  FACT Sales

  MEASURES OfferPrice, AgreedPrice, Status

  DIMENSIONS EstateID, OwnerID, CustomerID, AgentID,

  TimeID

- supervisors should be able to control the work of the agents by analyzing the visits to the estates, which the agents are in charge of

  FACT Viewing

  MEASURES Duration

  DIMENSIONS EstateID, CustomerID, AgentID, TimeID

Solution for Class Assignment (4)

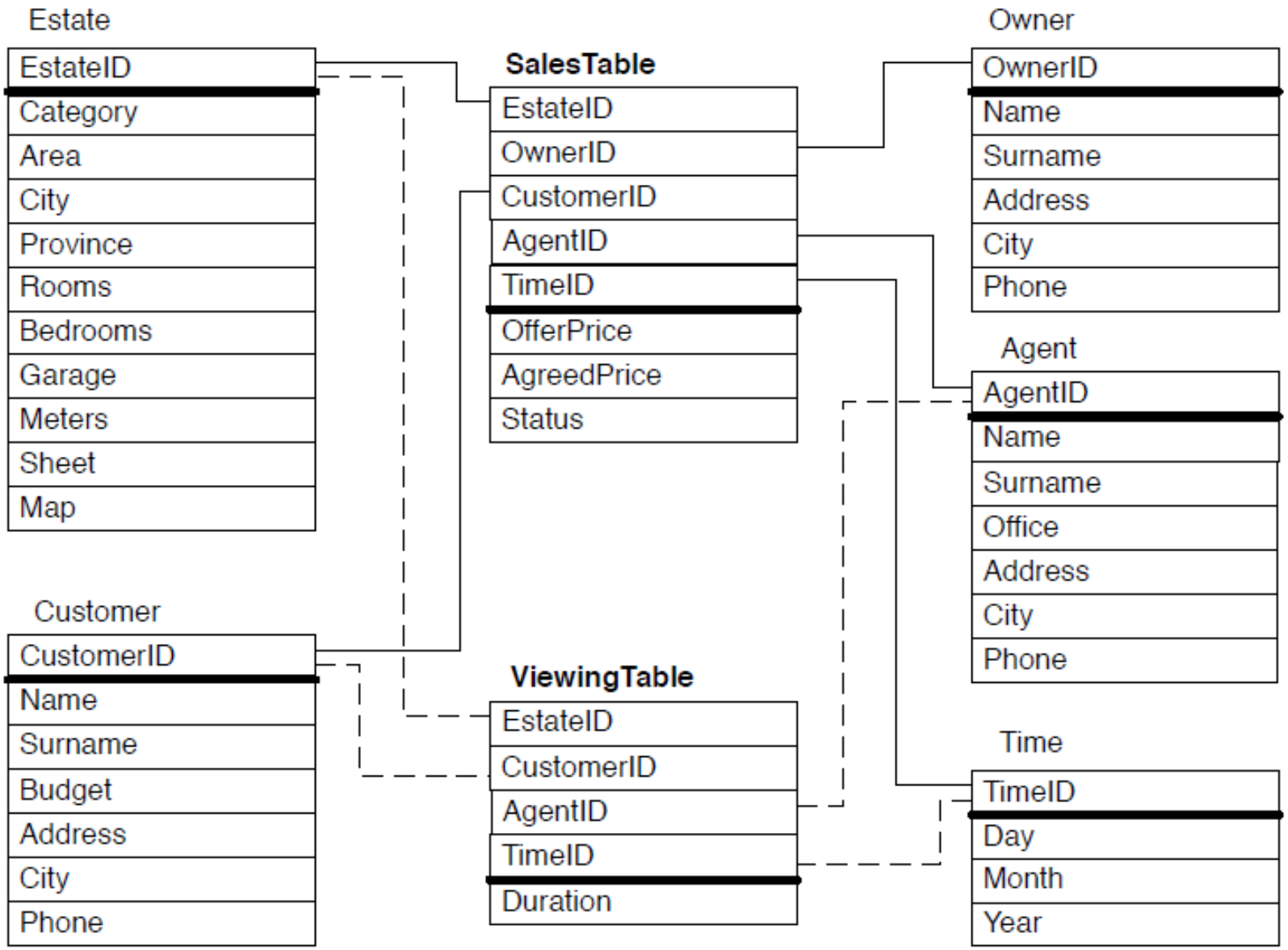


Figure: Star schema

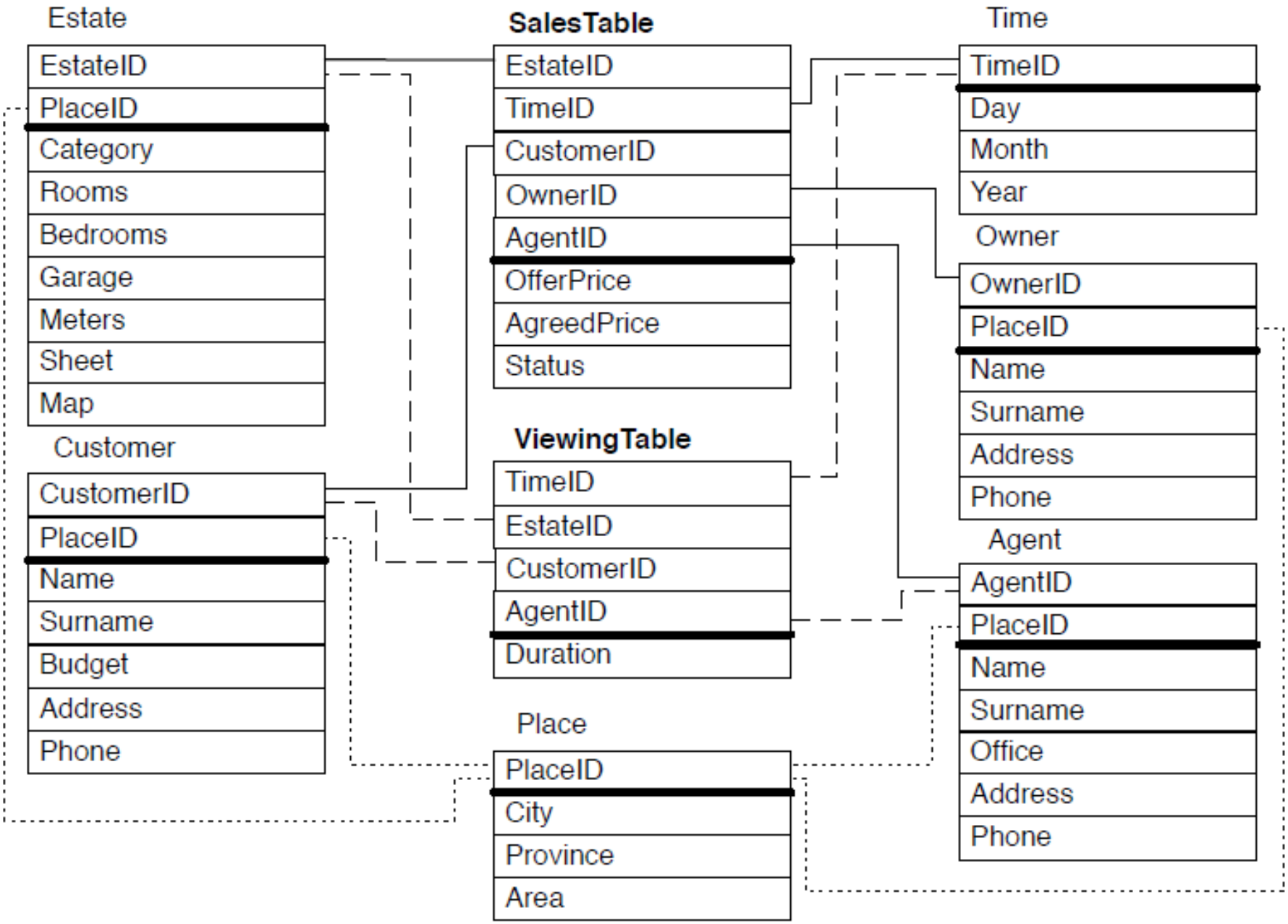


Figure: Snowflake schema

Class Assignment (5)

An online order wine company requires the designing of a data warehouse to record the quantity and sales of its wines to its customers. Part of the original database is composed by the following tables:

CUSTOMER (Code, Name, Address, Phone, BDay, Gender)

WINE (Code, Name, Type, Vintage, BottlePrice, CasePrice, Class)

CLASS (Code, Name, Region)

TIME (TimeStamp, Date, Year)

ORDER (Customer, Wine, Time, nrBottles, nrCases)

Note that the tables represent the main entities of the ER schema, thus it is necessary to derive the significant relationships among them in order to correctly design the data warehouse.

Construct Snowflake Schema.

Solution for Class Assignment (5)

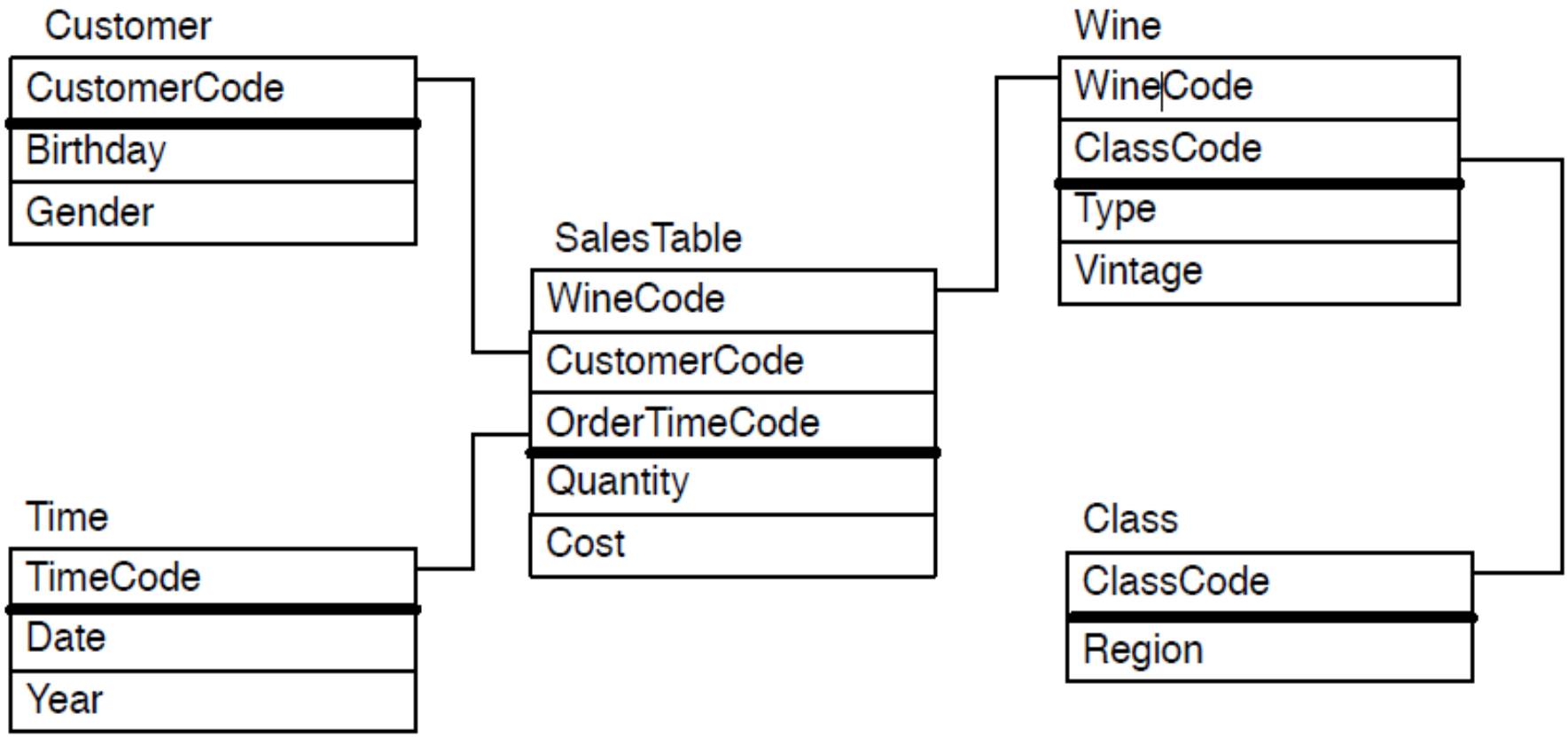


Figure: Snowflake schema

FACT Sales
MEASURES Quantity, Cost
DIMENSIONS Customer, Area, Time, Wine $\longrightarrow$ Class

References

1. Sam Anahory, Dennis Murray, “Data warehousing In the Real World”, Pearson Education.
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3. Teorey, T. J., “*Database Modeling and Design: The Entity-Relationship Approach”*, Morgan Kaufmann Publishers, Inc., 1990.
4. “An Overview of Data Warehousing and OLAP Technology”, S. Chaudhuri, Microsoft Research
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Thank you !!!